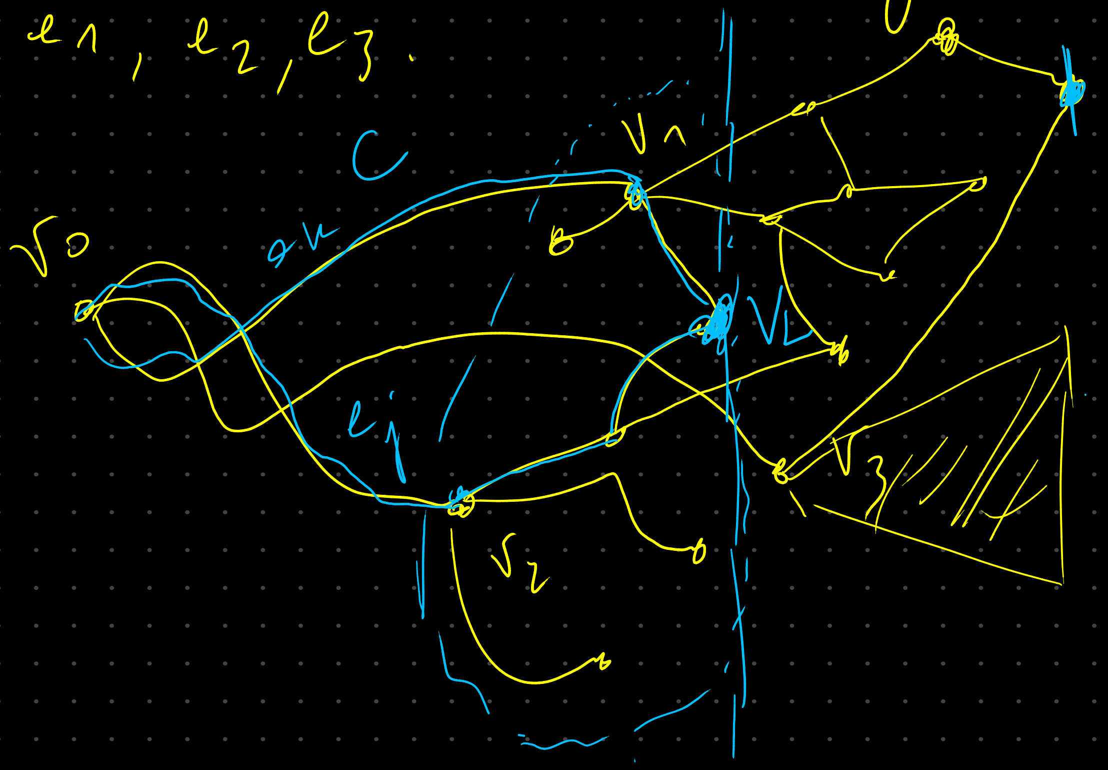
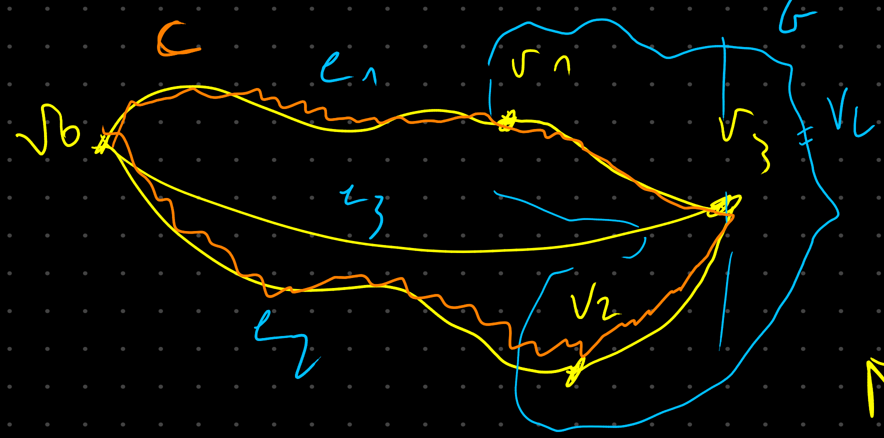
ruty: Case not 1

let v_L with $\pi(v_L)$ the smallest so that the subgraph induced by vertices v such that $\pi(v_0) < \pi(v) \leq \pi(v_L)$ has a component that contains at least two right endpoints of e_1, e_2, e_3.



$\{i, j, k\} = \{1, 2, 3\}$,

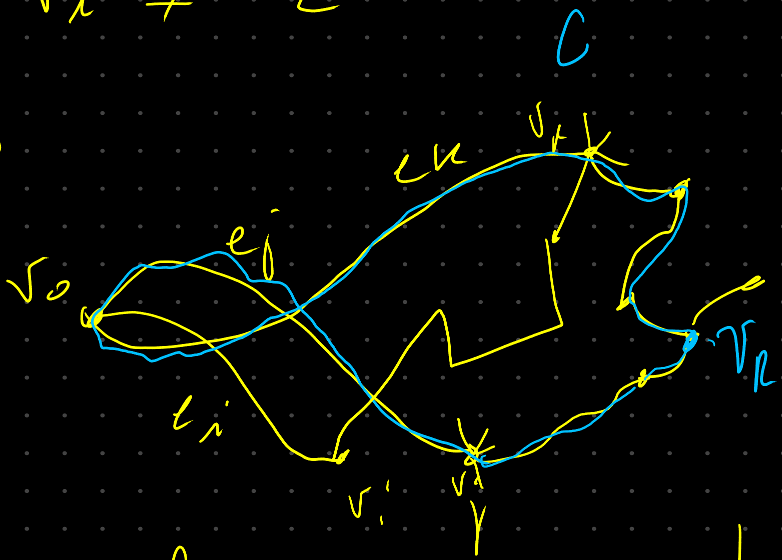


if $vv_L \in \{e_1, e_2, e_3\}$

we assume that $vv_L \in C$ so we can't have a cycle like this one.

$\Rightarrow v_i \neq v_L$

Lemma 5:



h_i' be comp of $h - V(C)$ that contains v_i.

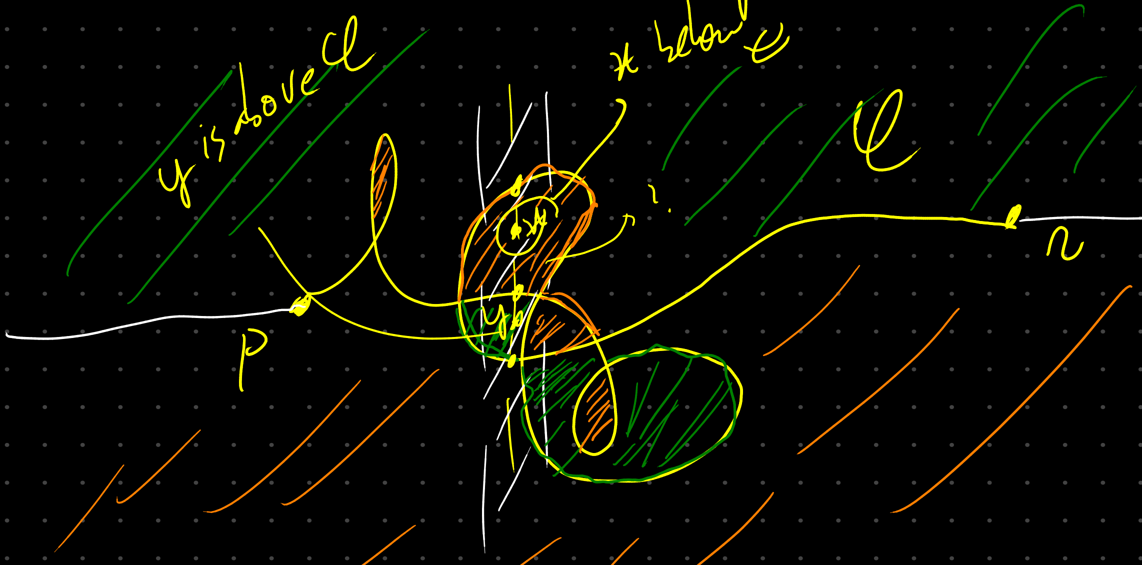
$\forall v$ of h_i' satisfies $x(v_0) < x(v) < x(v_R)$

[every int. edge moves evenly

– I think my issue is not understanding this new terminology and its use in the 2 lemmas.

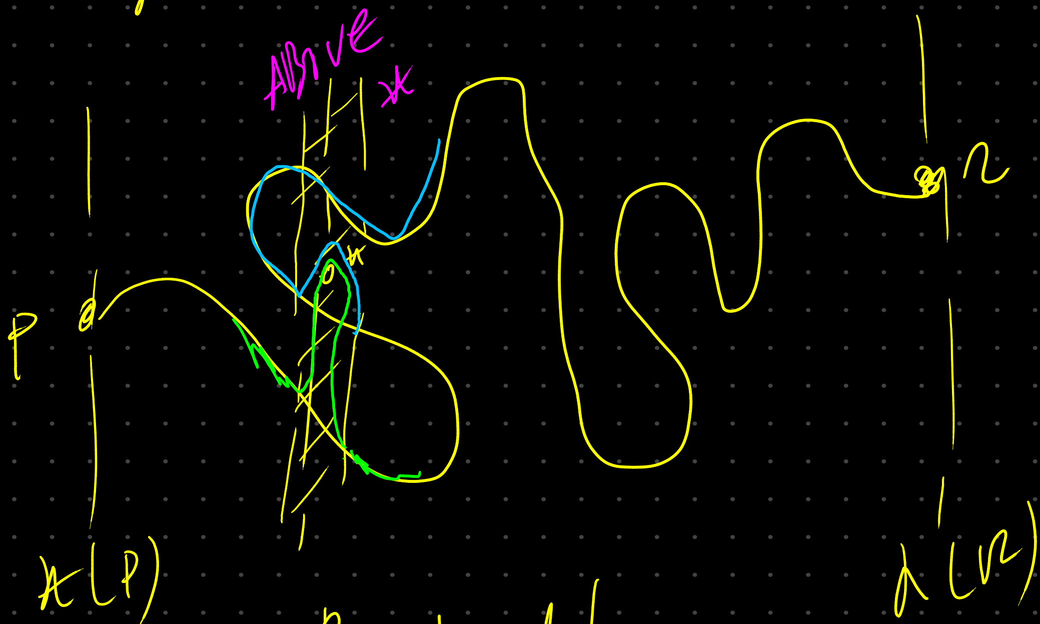
New terminology:

goal: generalization the notion of laying $\lessgtr$ above / below a curve to curves with self-intersection.



similar to the definition of a bounded edge

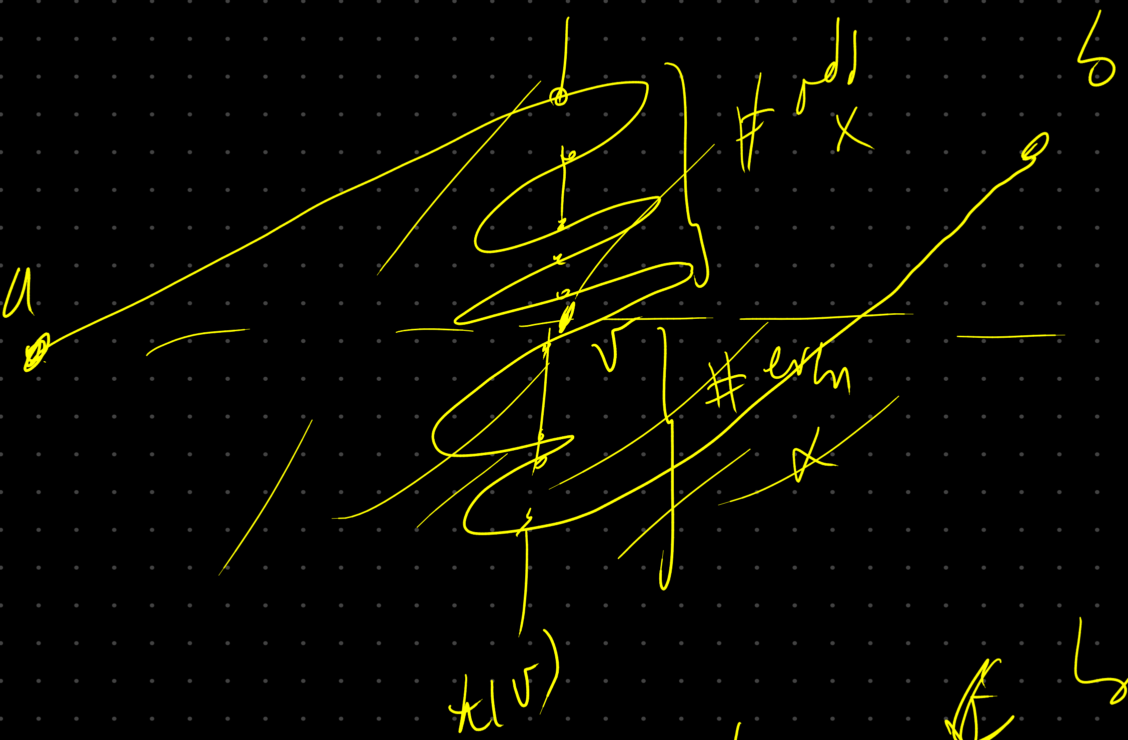
bounded edge:



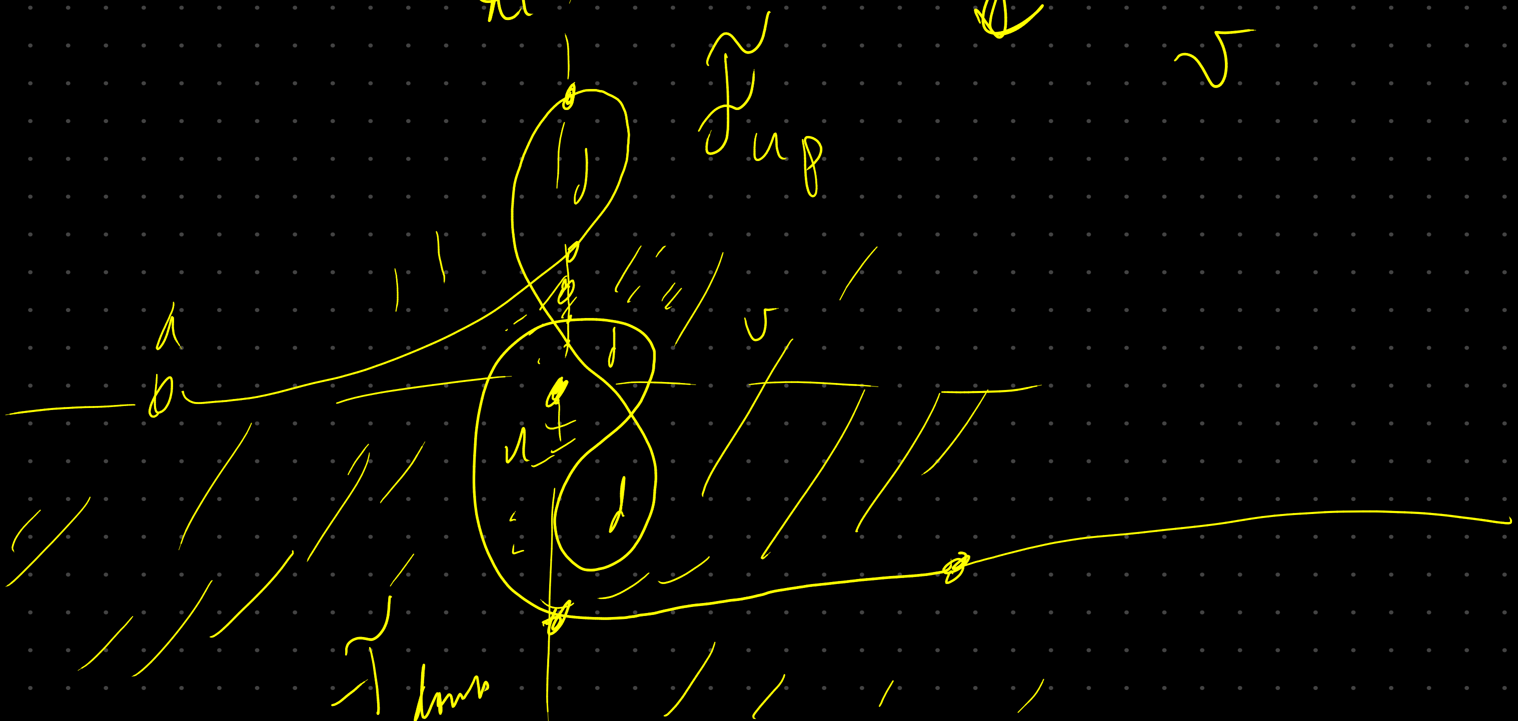
the purpose behind defining bounded edge is to weaken the constraints and we have lemma 2 stating that we can redraw bounded to x-monotone without changing rot and the parity of crossing between e and other edges of G in its proof the below/above the curve was introduced.

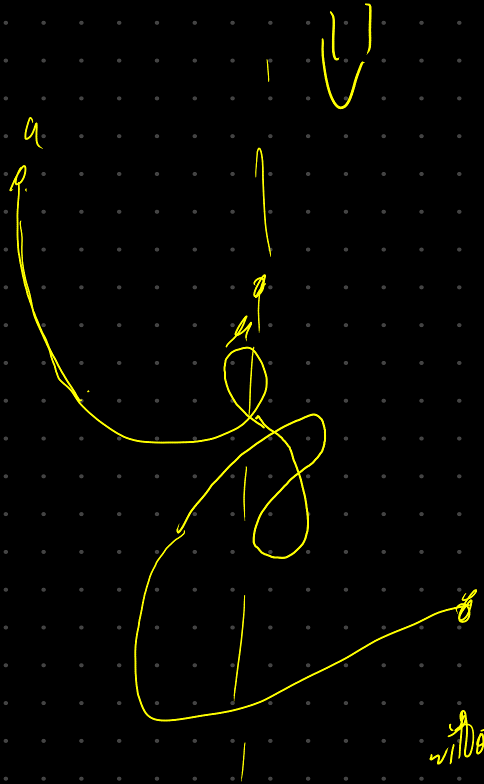
* going through proof lemma 2 again

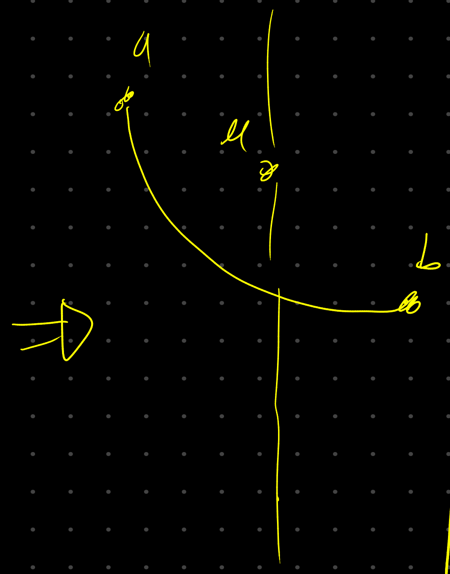
$e = ab$, $v \in V(H)$ arbitrary
$\star(a) < \star(v) < \star(b)$



$\mathcal{E}$ below v







without changing parity of crossings

~~it's just on the intersection the # of crossing of e_3 need to be even.~~

the new lemma. just extends the proof for self-intersect! (does it tho?)

— I think —

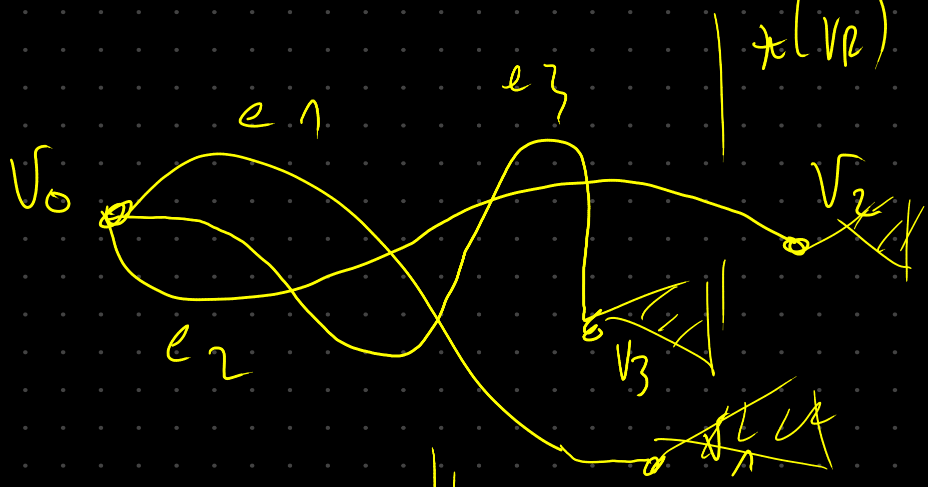
I just understood the usage of self-intersections.

$\Downarrow$

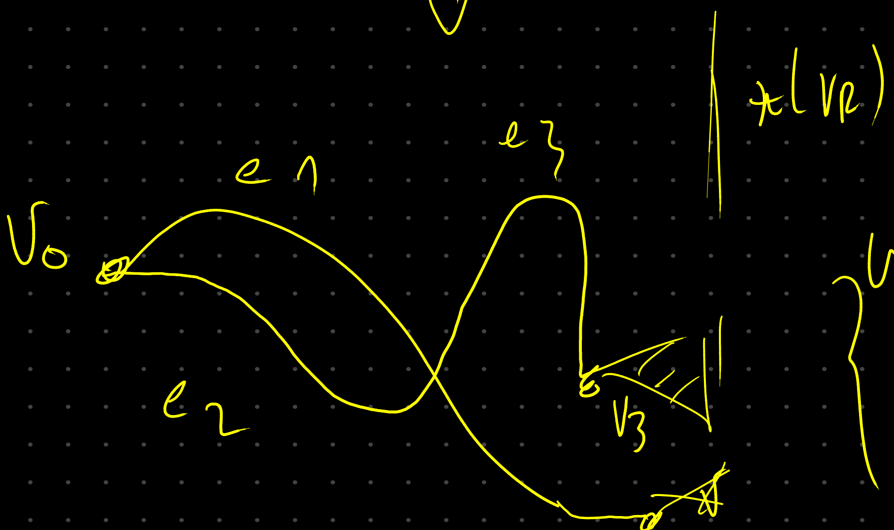
it's to introduce the edge e_3 that has even crossing with e_1 and e_2 while e_1, e_2 have an odd crossing !!!

not really, it's just a generalization

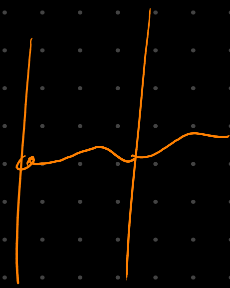
$v_k > \pi(v_0)$ define G' induced by G on vertices v with $\pi(v_0) < \pi(v) \leq \pi(v_k)$. Let G'_i be a comp. of G' that contains v_i (if $\pi(v_i) > \pi(v_k)$, then $G'_i = \emptyset$)



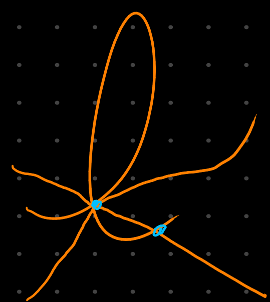
$\Downarrow$

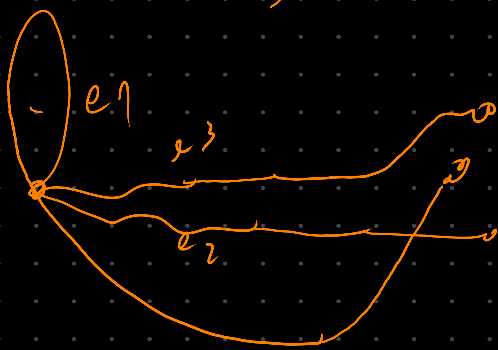


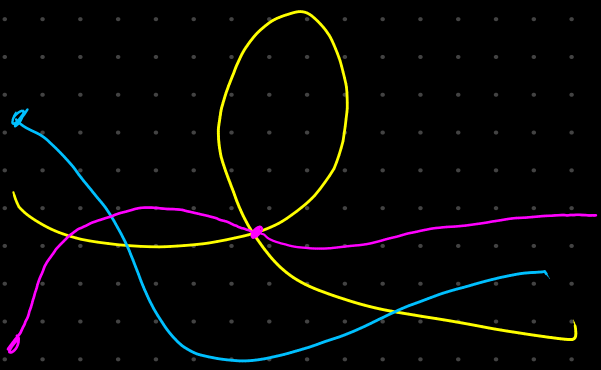
$$\begin{cases} G'_2 = \emptyset \\ G'_1 \neq \emptyset \\ G'_3 \neq \emptyset \end{cases}$$

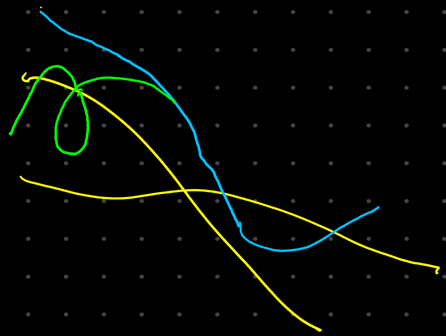


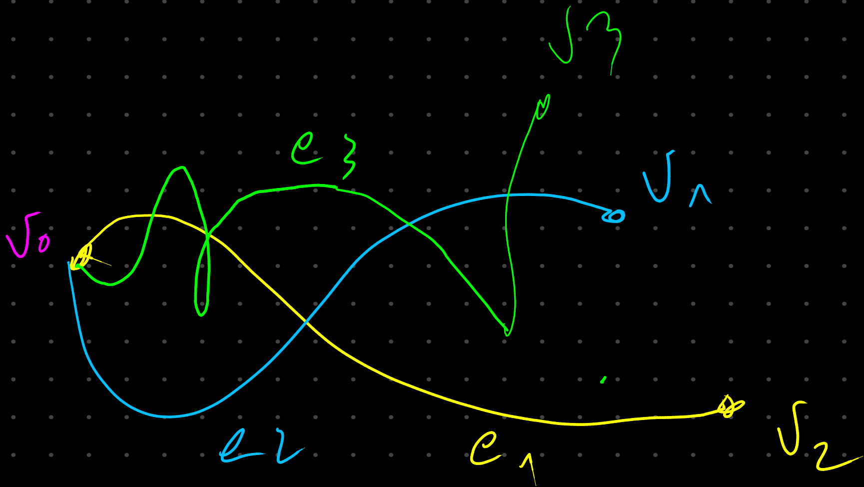
if not π-monotone drawing

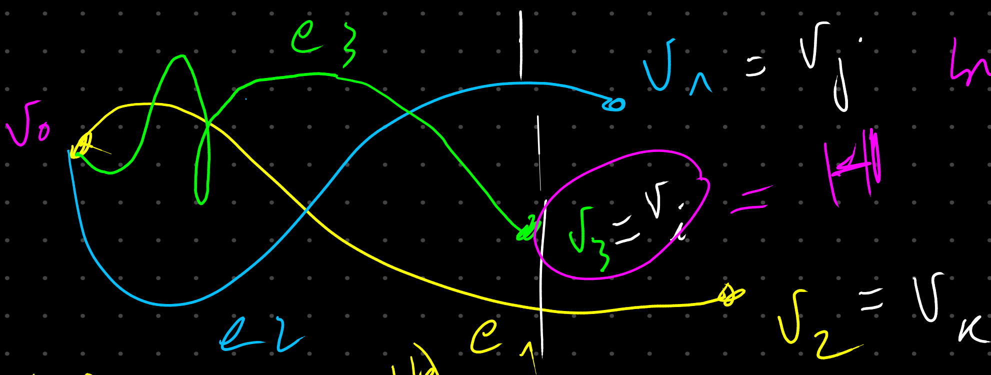












contradiction.

$\Uparrow$

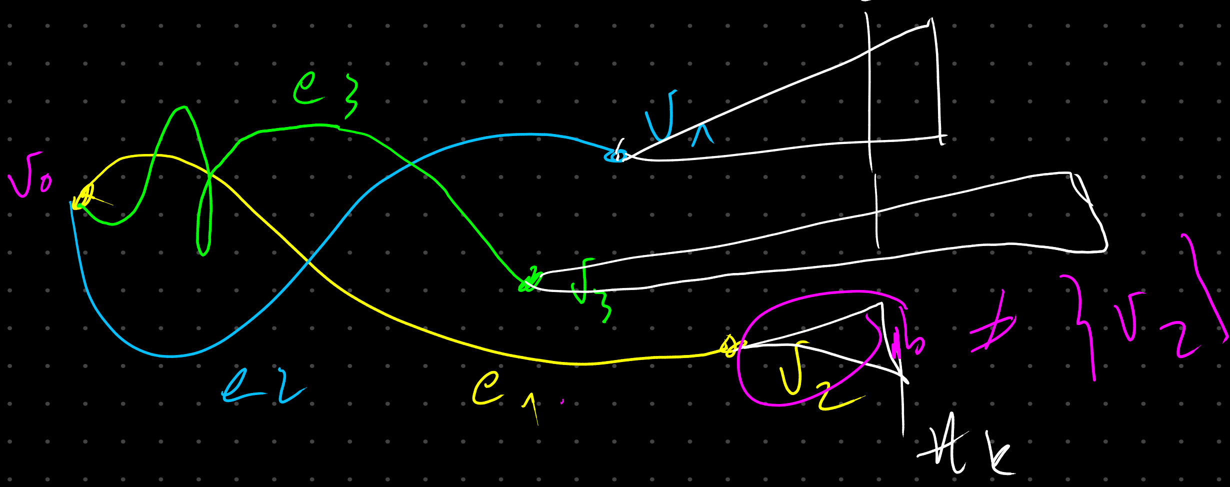
but we have $ac = v_0 v_1 \in E(H)$ with $\alpha(v_1) > \alpha(v_2)$

$\begin{cases} V(H) = \{b\} \\ \forall c \in V(H): ac \in E(H) \Rightarrow \alpha(c) \leq \alpha(b) \end{cases}$

t_k

$h'_j = \emptyset$

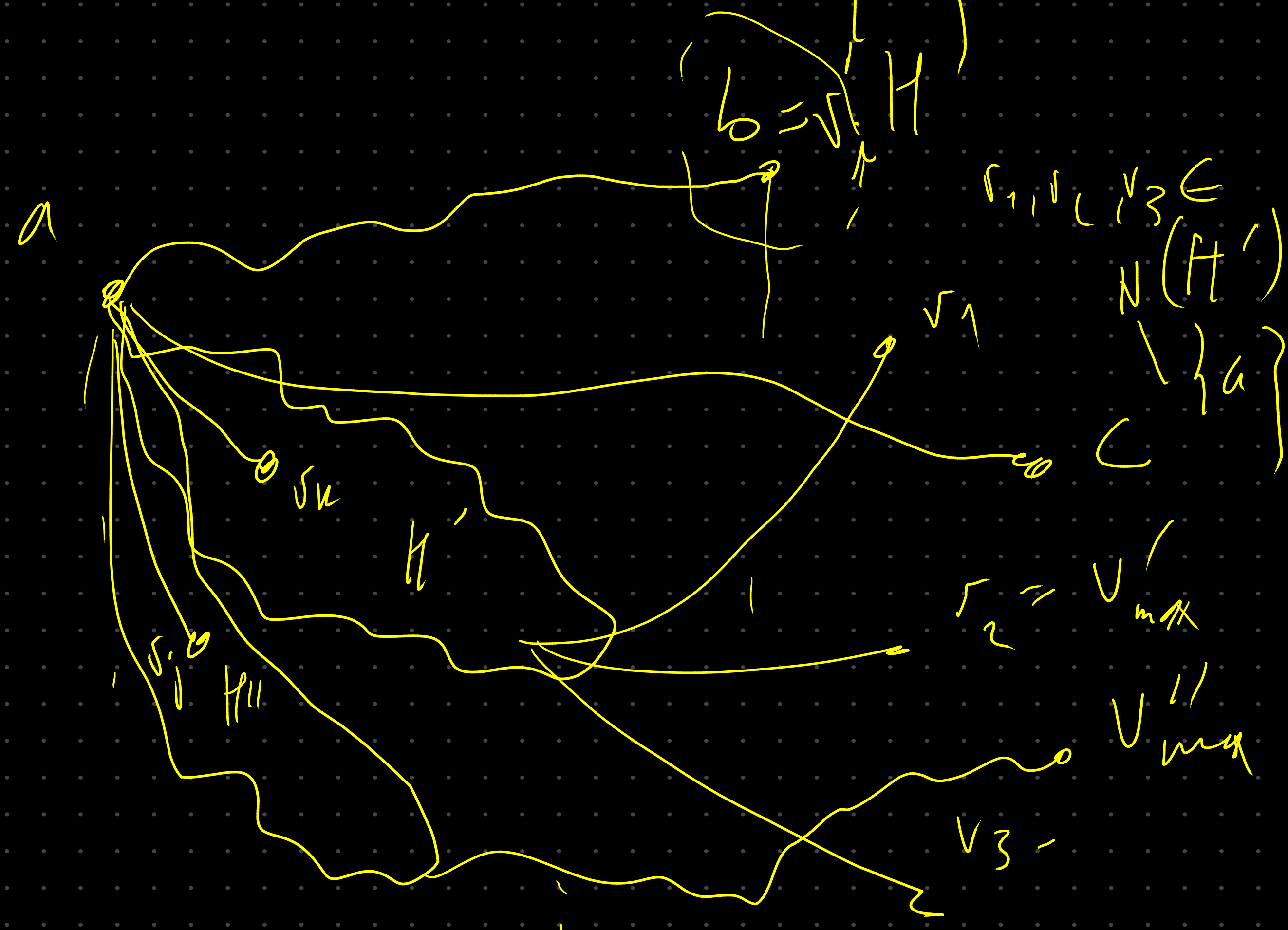
$h'_k = \emptyset$



$G'_i = V_i = V'_i$ $\begin{cases} G'_j \neq \emptyset \\ G'_k \neq \emptyset \end{cases}$

$\pi(v) \in \,]\pi(a), \pi(b)[\quad \forall v \in V(H'),\ a \in N(H') \neq \{a\}$

$\pi(v) > \pi(b) \quad \forall v \in N(H') \setminus \{a\}$



$\Downarrow$

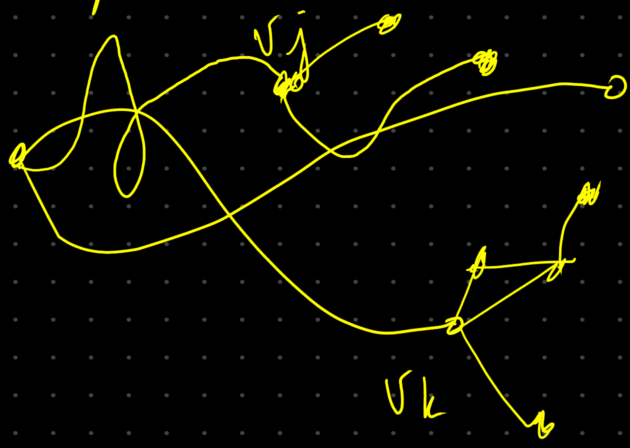
if we have 3 distinct components after removing a $\Rightarrow$ the variable 1 is a cut-vertex of G. By lemma 3 iii

if we have a comp. G'_i

it will have only one vertex,

and $\begin{cases} \#(v_j) \leq \#(v_i) \\ \#(v_k) \leq \#(v_i) \end{cases}$ ⟸ two ac with $\#(c) > \#(u)$

so now we need to consider v_j', v_k', if we use lemma 3 iii) second part then the induced subgraph can't be more than $\#(v_i')$

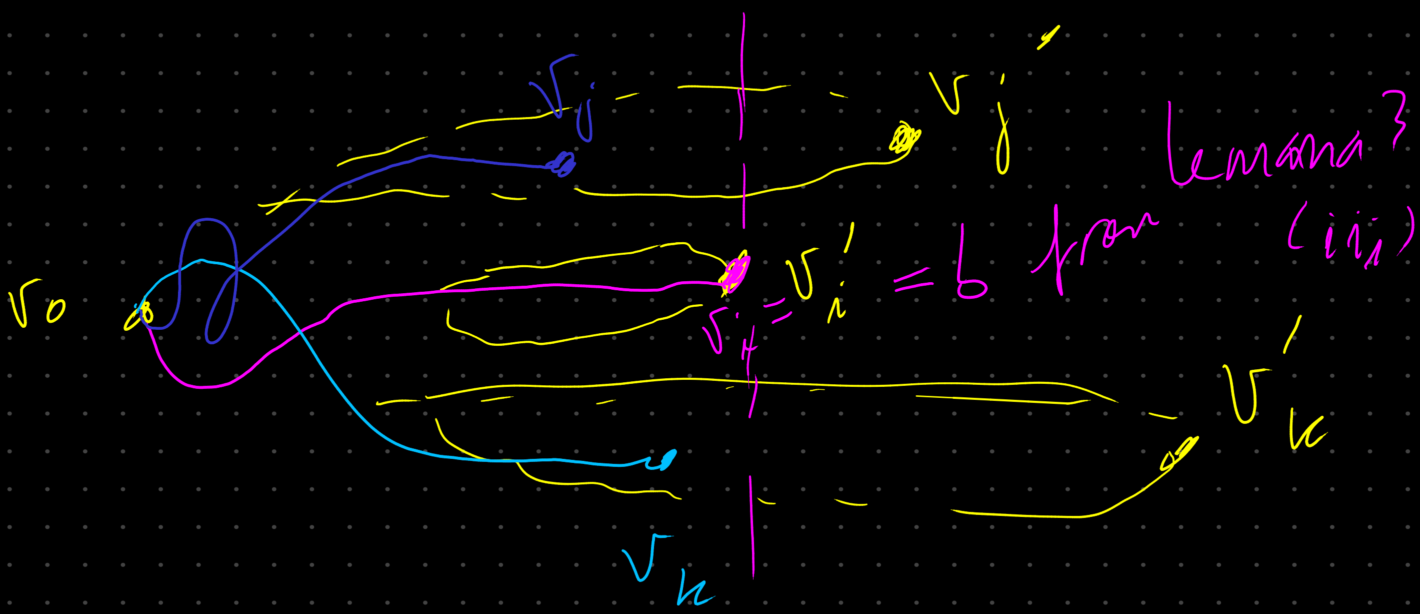


v_i has only one elem.

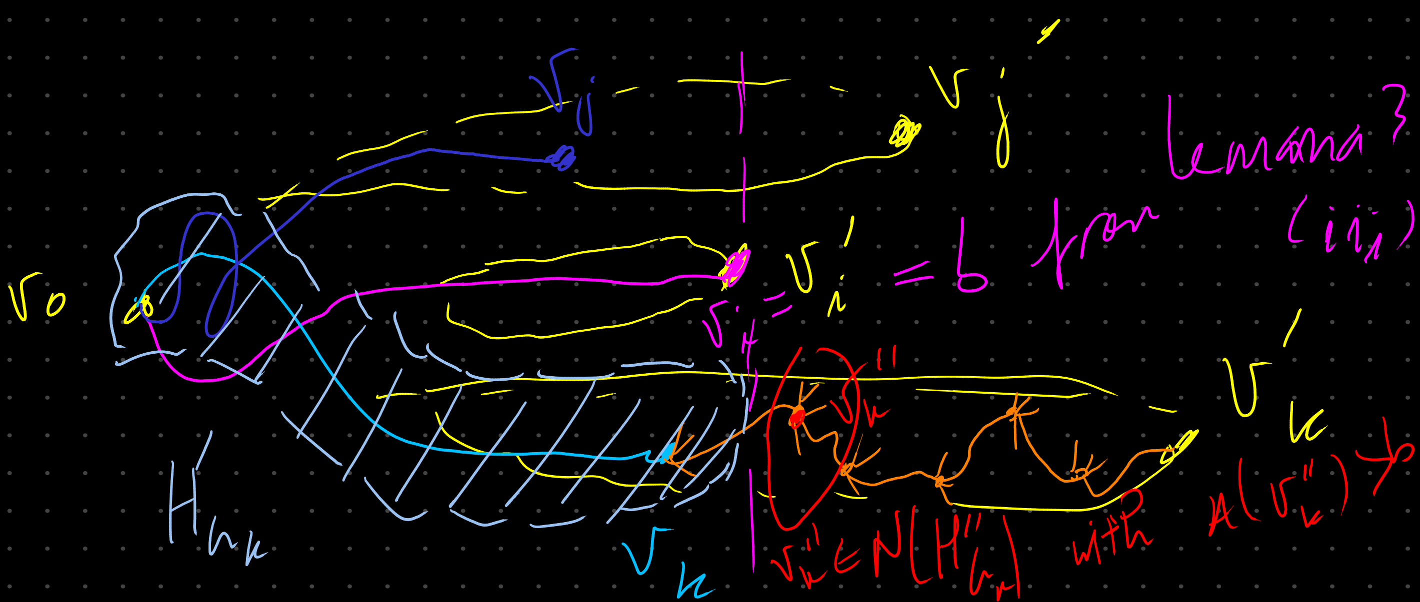
$\Rightarrow$ it means that the graph finishes in v_i

↓ step 2

↓ step 3



↓ step 4



contradiction because $\forall v \in N(H'_{v_k}) \setminus a$

$$\boxed{\pi(v) > \pi(v'_i) = \pi(L)}$$

Lemma 4

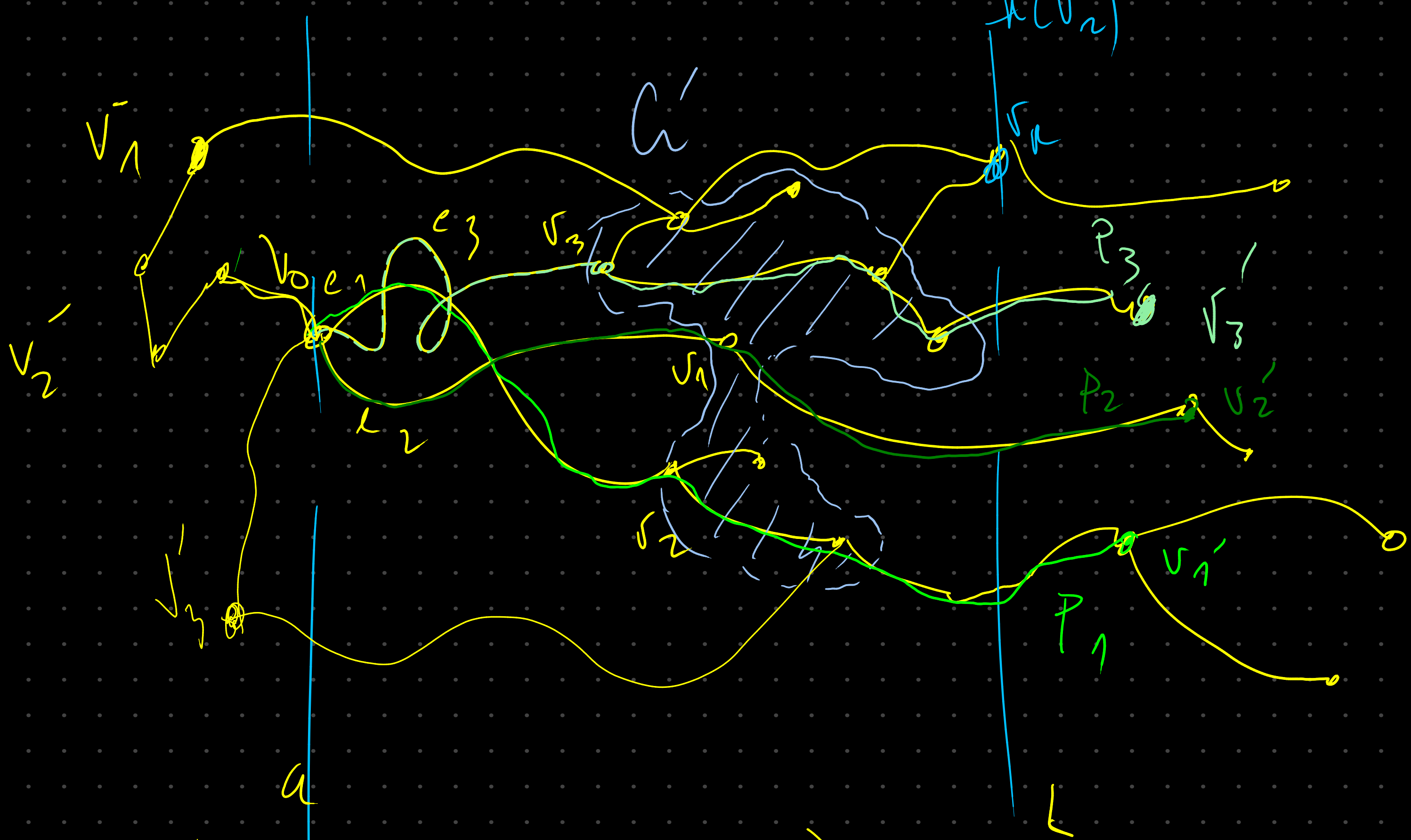
G' induced by G on vertices v with $\pi(v_0) < \pi(v) \leq \pi(v_R)$. G'_i comp. of G' that contains v_i

if: G'_1, G'_2, G'_3 pairwise disjoint

$\forall i \in \{1,2,3\}$ v_0 e_i

$\forall v \in P_i$ $\pi(v) \geq \pi(v_0)$

$\Rightarrow G'_i$ has no neighbors in G to the left of $\pi(v_0)$



false $\leftarrow N(H) = \{a, b\}$

true $\leftarrow V(G) \setminus (V(H) \sqcup \{a, b\}) \neq \emptyset$

cut vertex v_0 and $G - \{a\}$ has a comp. H with $\kappa(a) < \kappa(v)$ $\forall v \in V(H)$

$\Rightarrow$ H has one vertex